DIPAYAN SARKAR

PhD Researcher in Bioinformatics & Computational Biology

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RESEARCH SUMMARY

Currently pursuing a PhD in bioinformatics and computational biology, focused on deep learning methods for biological sequence-based prediction and generative model development. Experience includes neural network training, transfer learning with pretrained protein language models, and analysis of large-scale biological datasets. Current work involves autoregressive generative modeling of biological sequences, with ongoing exploration of diffusion-based approaches.

WORK EXPERIENCE

PhD Research Scholar
University of North Bengal, Dept. of Bioinformatics
2022 – Present | Siliguri, India

- Designed and implemented deep learning architectures for sequence-based protein–protein interaction prediction.
- Developed attention-based hybrid models integrating transfer learning from pretrained protein language models.
- Built and deployed web platforms for model inference and protein interaction network visualization.

PUBLICATIONS

1. Sarkar, D., & Sarkar, C. (2025). AttnSeq-PPI: Enhancing protein-protein interaction network prediction using transfer learning-driven hybrid attention. *BBA—Proteins and Proteomics*, 141102.
2. Sarkar, C.; Sarkar, D.; Parsad, R.; Mishra, D. (2022). Package `EGRNi`. CRAN.

SOFTWARE & TOOLS

AttnSeq-PPI
Deep learning-based PPI prediction platform — `compbiosysnbu.in/attnseqppi/`

ARACoFusion-PPI
Arabidopsis-specific PPI prediction tool — `aracofusion.compbiosysnbu.in/`

EDUCATION

Ph.D. Bioinformatics
University of North Bengal
Expected Mar 2027

- Advisor: Dr. Chiranjib Sarkar
- Focus: Deep learning for PPI network prediction

M.Sc. Botany
University of North Bengal
2021

Specialization: Biochemistry
77.25% (First Class)

B.Sc. (Hons.) Botany
Ananda Chandra College, UNB
2019

63.50% (First Class)

Higher Secondary (XII)
Haldibari High School, WBCHSE
2016

80.00% (First Class)

Secondary (X)
Haldibari High School, WBBSE
2014

91.58% (First Class)

SKILLS

Python | PyTorch | HuggingFace | R | C

Transformers | Attention Mechanisms | Transfer Learning | Sequence Modeling

Autoregressive Models | Diffusion Models

GPU Training | HPC | Bioinformatics

Docker | Flask | FastAPI | NumPy | Pandas | Matplotlib

AWARDS

CSIR-NET (JRF)
Life Sciences, June 2021 — AIR **216**

GATE
Life Sciences (XL), 2022